



Scan2BIM

i Purpose: This live doc summarizes the Scan2BIM automation project, its objective, core workflow, and expected outcomes for teams involved in digital construction documentation and BIM delivery.

Project Summary

The **Scan2BIM automation project** is designed to convert raw point cloud data into a structured **3D Building Information Model (BIM)** with minimal manual effort. Its goal is to improve the speed, consistency, and quality of construction documentation while making scanned building data easier to share and use across teams.

The solution ingests `.e57` point cloud files and processes them through a set of microservices. These services transform scan data into an **Industry Foundation Classes (IFC)** file that represents the scanned structure in a standardized, interoperable BIM format.

Business Goal

- Reduce manual modeling effort from scan data
- Accelerate delivery of accurate as-built documentation
- Improve collaboration across design, engineering, and construction teams
- Provide a repeatable and scalable pipeline for BIM generation

High-Level Workflow

1. **Input ingestion:** Raw building scan data is provided as `.e57` files.
2. **Automated processing:** A series of microservices parse, validate, and transform the point cloud data.

- 3. **Model generation:** Structural and spatial information is interpreted and assembled into a BIM representation.
- 4. **Output delivery:** The pipeline produces an IFC file representing the scanned building.

Core Architecture

Component	Role	Outcome
.e57 Ingestion	Receives and prepares raw point cloud data for processing	Standardized input for downstream services
Microservices Pipeline	Executes modular transformation and processing steps	Automated, scalable workflow orchestration
IFC Generation	Builds a BIM-compatible model from processed scan data	Interoperable 3D building model output

Expected Benefits

✔ **Key value:** By automating the journey from scan to BIM, the project supports faster documentation, more reliable model generation, and better information sharing across project stakeholders.

- **Efficiency:** Less manual modeling and rework
- **Consistency:** Standardized processing through microservices
- **Interoperability:** IFC output supports broad BIM tool compatibility
- **Collaboration:** Easier sharing of accurate building data across teams

▼ Scope at a glance

The project focuses on automated transformation of point cloud scan data into BIM deliverables. It is centered on ingesting .e57 files, processing them through distributed services, and producing IFC output suitable for downstream construction and documentation use cases.